

```
1  <span class="keyword">#include</span> <span class="header-file"><stdio.h></span>
2  <span class="keyword">#include</span> <span class="header-file"><stdlib.h></span>
3
4  int main() {
5      int n, head, seek_time = 0;
6
7      printf("Enter the number of disk requests: ");
8      scanf("%d", &n);
9      int request_queue[n];
10
11     printf("Enter the disk request queue:\n");
12     for (int i = 0; i < n; i++) {
13         scanf("%d", &request_queue[i]);
14     }
15
16     printf("Enter the initial position of the disk head: ");
17     scanf("%d", &head);
18
19     // FCFS Scheduling
20     printf("\nFCFS Disk Scheduling:\n");
21     printf("Head Movement Sequence: %d", head);
22     for (int i = 0; i < n; i++) {
23         seek_time += abs(head - request_queue[i]);
24         head = request_queue[i];
25         printf(" -> %d", head);
26     }
27
28     printf("\nTotal Seek Time: %d\n", seek_time);
29     printf("Average Seek Time: %.2f\n", (float) seek_time / n);
30
31     return 0;
32 }
```

```
Enter the number of disk requests: 5
Enter the disk request queue:
82 170 43 140 24
Enter the initial position of the disk head: 50

FCFS Disk Scheduling:
Head Movement Sequence: 50 -> 82 -> 170 -> 43 -
> 140 -> 24
Total Seek Time: 460
Average Seek Time: 92.00

=== Code Execution Successful ===
```